

Module 2.2: Data Collection and Analysis

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Hypothesis Testing

Hypothesis testing is a **statistical method** used to determine whether there is enough evidence to reject a given assumption (hypothesis) about a population.

1. Null Hypothesis (H_0)

- The **default assumption** that there is **no significant effect or difference** between 2 variables in a hypothesis.
- Example: "There is no difference in test scores between students who study at night and those who study in the morning."

2. Alternative Hypothesis (H_1 or H_a)

- The opposite of the null hypothesis; it suggests that **there is a significant effect or difference** between 2 variables in a hypothesis.
- Example: "Students who study at night perform better than those who study in the morning."

3. Type Error

- a. **Type I Error (False Positive):** Rejecting the **null hypothesis** when it is actually **true**.
- b. **Type II Error (False Negative):** **Failing to reject** the null hypothesis when it is actually **false**.

A **man is innocent**, but the court **finds him guilty**.

- **Null hypothesis (H_0):** He is innocent.
- **Reality:** He is innocent.
- **Court's decision:** Rejects $H_0 \rightarrow$ Says guilty. **TYPE I ERROR**

A **criminal is guilty**, but the court **lets him go free**.

- **Null hypothesis (H_0):** He is innocent.
- **Reality:** He is guilty.
- **Court's decision:** Fails to reject $H_0 \rightarrow$ Says innocent. **TYPE II ERROR**

4. Significance Levels (α)

- The probability of **rejecting a true null hypothesis** (Type I error).
- Common significance levels:
 - **0.05 (5%)** $\rightarrow$ Most commonly used.
 - **0.01 (1%)** $\rightarrow$ Stricter, used in high-precision research.
 - **0.10 (10%)** $\rightarrow$ Less strict, used in exploratory research.

5. P-value Interpretation

- The **P-value** indicates the probability of obtaining test results **at least as extreme as the observed data**, assuming the null hypothesis is true.
- **Interpretation:**
 - $P \leq \alpha$ (e.g., 0.05) $\rightarrow$ **Reject H_0** $\rightarrow$ Significant result (strong evidence for H_1).
 - $P > \alpha \rightarrow$ **Fail to reject H_0** $\rightarrow$ No significant evidence to support H_1 .

Example of Hypothesis Testing:

Hypothesis: "A new drug improves recovery time compared to an old drug."

- **H_0 :** The new drug has no effect on recovery time.
- **H_1 :** The new drug improves recovery time.
- If **$P = 0.03$** (less than 0.05), we **reject H_0** and conclude that the new drug significantly improves recovery.

Data Analysis with Statistical Packages

1. SigmaStat

- A statistical analysis software that helps with hypothesis testing and validation.
- Features include:
 - T-tests
 - ANOVA
 - Regression analysis
 - Non-parametric tests

2. SPSS (Statistical Package for the Social Sciences)

- A widely used software for statistical analysis in research.
- Provides both **descriptive and inferential statistics**.

Key Statistical Tests in SPSS

a) Student t-test: Used to compare the **means of two groups**. Types:

- **Independent t-test:** Compares two different groups.
- **Paired t-test:** Compares the same group before and after an event.

b) ANOVA (Analysis of Variance): Used to compare the means of three or more groups. Types:

- **One-way ANOVA:** Tests one independent variable.
- **Two-way ANOVA:** Tests two independent variables.

c) Chi-square Test

- Used to test relationships between **categorical variables**.
- Determines if the observed distribution differs from the expected distribution.

d) Regression Analysis: Determines the relationship between **independent and dependent variables**.

Types:

- **Linear Regression:** Predicts outcomes using a straight-line relationship.
- **Multiple Regression:** Uses multiple independent variables to predict an outcome.

Chi-Square Test

The Chi-Square (χ^2) Test is a non-parametric statistical test used to determine if there's a significant association between two categorical variables or if observed frequencies differ from expected frequencies.

- Data should be **frequencies**, not percentages.
- Categories should be **mutually exclusive**.
- Expected frequency in each cell should be ≥ 5 (for validity).
- Observations are **independent**.

Large Sample Test/Z test

Large sample tests are used when the sample size is **30 or more** ($n \geq 30$).

Steps to Perform a Z-Test

1. **State Hypotheses:**
 - Null hypothesis (H_0): No difference or claim is true
 - Alternative hypothesis (H_1): What you want to test
2. **Choose Significance Level (α):** Usually 0.05 or 0.01
3. **Calculate Test Statistic (Z):** Use appropriate formula
4. **Determine Critical Value:** Based on one-tailed or two-tailed test
5. **Make a Decision:**
 - If $|Z| > Z_{\text{critical}}$, **reject H_0**
 - Else, **do not reject H_0**

Small Sample Test/t test/student t test

A **small sample test** is used when the **sample size is less than 30** and the **population standard deviation (σ) is unknown**.

- Since the sample is small, we can't rely on the **Central Limit Theorem**.
- Instead of the Z-distribution, we use the **Student's t-distribution**.

Steps to Conduct a Small Sample t-Test

1. **Formulate Hypotheses**
 - H_0 : No difference
 - H_1 : There is a difference
2. **Choose Significance Level (α):** Usually 0.05 or 0.01
3. **Compute t-statistic** using the correct formula
4. **Find Critical t-value** from **t-table** using:
 - Degrees of freedom (df):
 - One-sample: $n-1$
 - Paired: $n-1$
 - Independent: n_1+n_2-2
5. **Make a Decision**
 - If $|t| > t_{\text{critical}}$, **Reject H_0**
 - Otherwise, **Fail to reject H_0**

Aspect	Large Sample Test	Small Sample Test
Sample Size	$n \geq 30$	$n < 30$
Population SD (σ)	Known or approximated	Unknown
Central Limit Theorem	Applicable	Not fully reliable
Test Used	Z-test	t-test
Accuracy	More reliable due to large data	Less reliable, needs careful handling
Variance Estimation	From population	From sample

Large Sample Test

$$Z\text{-test by } (z) = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$z < z_{\text{given}}$ then H_0 accept.

$\bar{x}$ = sample mean
 μ = pop mean
 n = sample size
 σ = pop s.d

Small Sample Test

$$t\text{-test by } t = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$t < t_{\text{given}}$, H_0 accept.

Chi-Square Test

$$f_e = \frac{\sum f}{N}$$

$$\chi^2_t = \frac{\sum (f_o - f_e)^2}{f_e}$$

f_e = expected freq.
 f_o = given freq.

χ^2 will be given

$\chi^2_t < \chi^2_{\text{given}}$, H_0 accepted

ANOVA (Analysis of Variance)

ANOVA is a statistical method used to test whether there are significant differences between the means of three or more independent groups.

- It generalizes the t-test (which compares only two means).
- Instead of comparing each pair individually (which increases error), ANOVA tests all groups simultaneously.

Steps in One-Way ANOVA

- State hypotheses**
 - $H_0: \mu_1 = \mu_2 = \mu_3$ (all means are equal)
 - H_1 : At least one mean is different
- Calculate group means and overall mean**
- Calculate:**
 - SSB** (Sum of Squares Between)
 - SSW** (Sum of Squares Within)
 - DF** (Degrees of Freedom)
 - MSB, MSW**, and then **F**

$$SSB = n \cdot [(\bar{X}_A - \bar{X}_{GM})^2 + (\bar{X}_B - \bar{X}_{GM})^2 + (\bar{X}_C - \bar{X}_{GM})^2]$$

$$SSW = \sum (x_i - \bar{X}_{\text{group}})^2 \text{ for all data}$$

for each group it is needed to be found then adder

F-Ratio Formula: $F = \frac{\text{Mean Square Between Groups (MSB)}}{\text{Mean Square Within Groups (MSW)}}$

- Compare:** **F value** with F-critical from F-table (based on degrees of freedom)
- Decision:** If $F_{\text{calculated}} > F_{\text{critical}}$, **reject H_0**

Q. Group A : 70, 72, 74
 Group B : 65, 67, 69
 Group C : 80, 82, 84

Ans. $\bar{X}_A = \frac{70+72+74}{3} = 72$ $\bar{X} = \frac{440}{9} = 77$
 $\bar{X}_B = \frac{65+67+69}{3} = 67$
 $\bar{X}_C = \frac{80+82+84}{3} = 82$

① Sum of Square btw grps.

$$SSB = n((\bar{X}_A - \bar{X})^2 + (\bar{X}_B - \bar{X})^2 + (\bar{X}_C - \bar{X})^2)$$

$$= 3.[5^2 + 10^2 + 5^2] = 450$$

② Sum of Square within grps.

$$SSW = (70-72)^2 + (72-72)^2 + (74-72)^2 + (65-67)^2$$

$$+ (67-67)^2 + (69-67)^2 + (80-82)^2 + (82-82)^2 + (84-82)^2$$

$$= 4 + 0 + 4 + 4 + 0 + 4 + 4 + 0 + 4$$

$$= 24$$

③ Degree of freedom.

$$DF_B = k-1 = 3-1 = 2$$

$$DF_W = N-k = 9-3 = 6$$

$$F = \frac{MSB}{MSW} = \frac{SSB/DF_B}{SSW/DF_W} = \frac{450/2}{24/6} = \frac{225}{4} = 56.25$$

④ Now from F-table, at DF_B , DF_W , $\alpha = 0.05$.

$$F_{\text{given}} = 5.14$$

$F > F_{\text{given}} \therefore \rightarrow \text{Reject } H_0$